

Introduction to NLP

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What is Natural Language Processing?

- **Natural language processing (NLP)** is a field of computer science, artificial intelligence concerned with the interactions between computers and human (natural) languages, and, in particular, concerned with programming computers to fruitfully process large natural language data.
- Natural language processing is concerned with the development of computational models of aspects of human language processing.
- Natural Language processing is an interdisciplinary field with many names such as **Speech and Language Processing, human language technology, natural language processing, computational linguistics, speech recognition and synthesis**

NLP Tasks/Applications

Core Tasks



Text
Classification



Information
Extraction



Conversational
Agents



Information
Retrieval



Question
Answering

General Applications



Spam
Classification



Calendar Event
Extracton



Personal
Assistants



Search
Engines

JEOPARDY!

Jeopardy!

Industry specific



Social Media
Analysis



Retail Data
Extraction



Health Records
Analysis



Financial
Analysis



Legal Entity
Extraction

NLP Tasks/Applications (Contd....)

Core Tasks:

- *Language Modeling*: task of predicting what the next word in a sentence will be based on the history of previous words.

Language modeling is useful for wide variety of problems such as speech recognition, optical character recognition, handwriting recognition, machine translation, and spelling correction.

- *Text Classification*: task of bucketing text into a known set of categories based on its content.

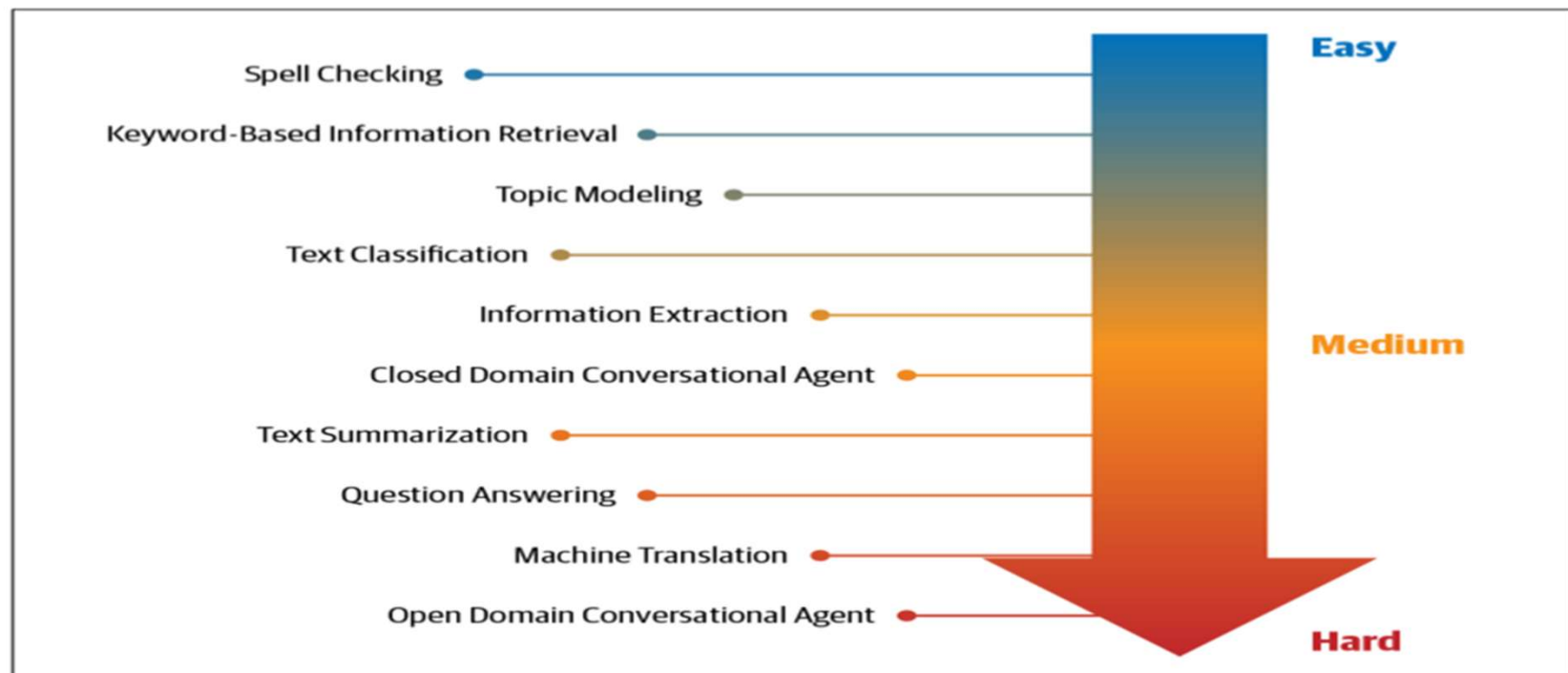
Text classification is used in a variety of tools from email spam classification to sentiment analysis.

- *Information Extraction*: extracting relevant information from text, such as calendar events from emails or names of people mentioned in a social media post.

NLP Tasks/Applications (Contd....)

- *Information Retrieval*: task of finding documents relevant to a user query from a large collection (example: google search)
- *Conversational Agent*: task of building dialogue systems that can converse in human languages. (example: Alexa, Siri, etc.)
- *Text Summarization*: task to create short summaries of larger documents while retaining the core content and preserving overall meaning of text.
- *Question Answering*: task of building a system that can automatically answer questions posed in natural language.
- *Machine Translation*: task of translating text from one language to another.
- *Topic Modeling*: task of uncovering the topical structure of a large collection of documents.

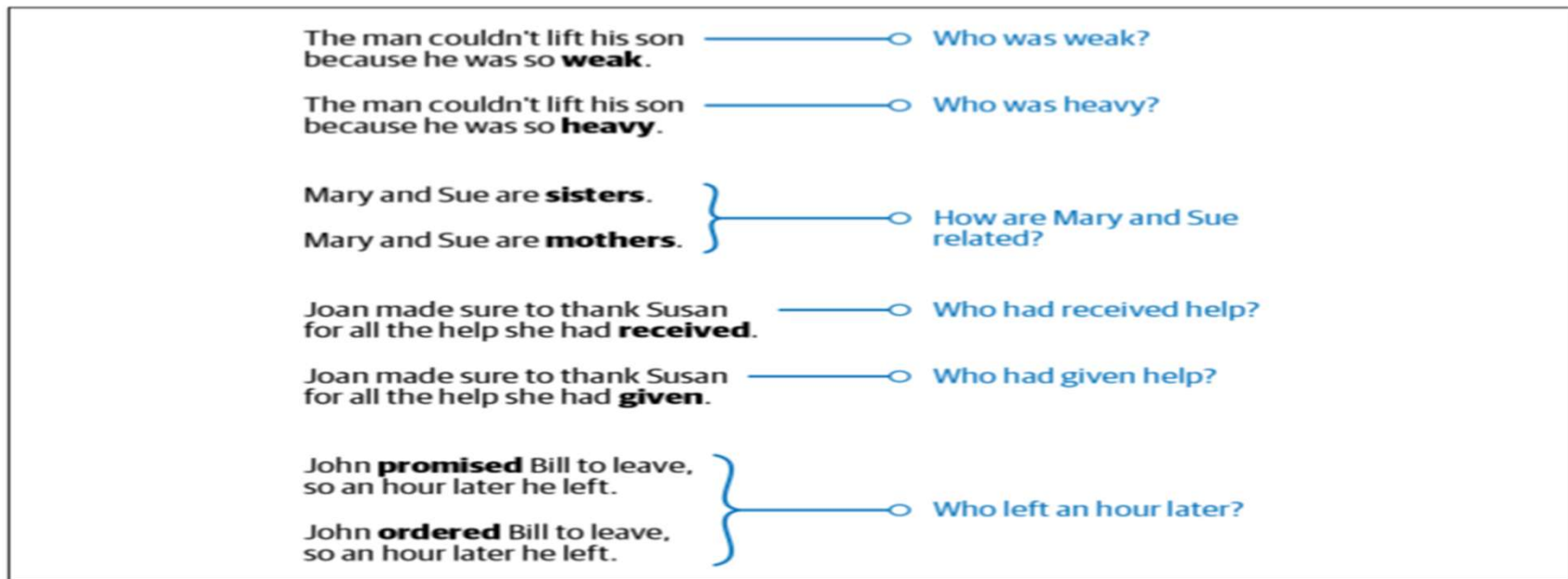
NLP Tasks/Applications (Contd....)



NLP tasks organized according to their relative difficulty

Why is NLP Challenging?

1. *Ambiguity*: Ambiguity means uncertainty of meaning.



Examples of ambiguity in language from the Winograd Schema Challenge

NLP Challenges (Contd....)

2. Common Knowledge:

- A key aspect of human knowledge is “common knowledge” i.e. set of all facts that human are aware of.
- In any conversation, it is assumed that these facts are known, hence they are not explicitly mentioned, but they do have bearing on the meaning of sentence.
- For example, consider two sentences: “man bit dog” and “dog bit man”

The two sentences are syntactically very similar, but a computer would find it very difficult to differentiate between the two, as it lacks the common knowledge humans have.

NLP Challenges (Contd....)

3. *Creativity*

- Language is not just rule driven; there is also a creative aspect to it such as various styles, dialects, genres, and variations are used in language.
- Poems are a great example of creativity in language.
- Making machines understand creativity is a hard problem not just in NLP; but in AI in general.

NLP Challenges (Contd....)

4. *Diversity across languages:*

- For most of the languages, there is no direct mapping between the vocabularies of any two languages.
- This makes porting an NLP solution from one language to another hard. A solution that works for one language might not work at all for another language.
- This means that one either builds a solution that is language independent or that one needs to build separate solutions for each language. The first solution is conceptually very hard, the second solution is laborious and time consuming.

Approaches to NLP

Different approaches used to solve NLP problems commonly fall into three categories:

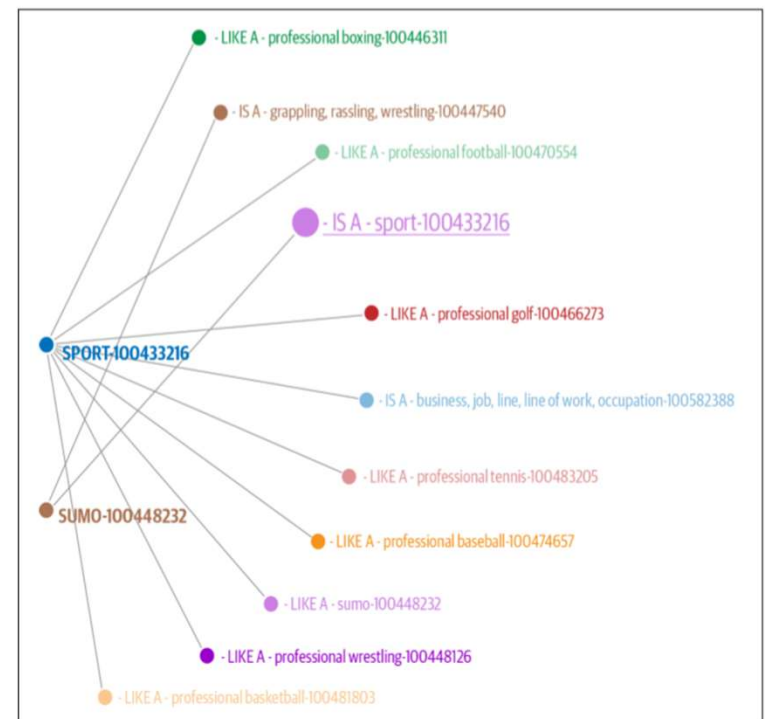
1. Heuristics-Based NLP
2. Machine Learning-Based NLP
3. Deep Learning-Based NLP

Heuristics-Based NLP

- Similar to other early AI systems, early attempts at designing NLP systems were based on *building rules for the task at hand*.
- This required the developers had some *expertise in the domain* to formulate the rules that could be incorporated into a program.
- These systems also required resources like *dictionaries and thesauruses*, typically compiled and digitized over a period of time.
- An example of solving a NLP task using resources is *lexicon-based sentiment analysis*. It uses counts of positive and negative words in the text to deduce the sentiment of text.

Heuristics-Based NLP (Contd...)

- Besides dictionaries and thesauruses, more elaborative knowledge bases have been built to aid NLP tasks such as *WordNet*.
- *Wordnet* is a database of words and the semantic relationships between them such as synonyms, hyponyms, meronyms, etc.
- Synonyms refer to words with similar meanings.
- Hyponyms capture is-type-of relationships. For example, baseball, sumo, wrestling are all hyponyms of sports.
- Meronyms capture is-part-of relationship. For example, hands and legs are meronyms of body.



-- Wordnet graph for the word "sport" [8]

Heuristics-Based NLP (Contd...)

- Dictionaries, thesauruses, WordNet are based on word-level information, rule-based systems go beyond words and can incorporate other forms of information.
- *Regular expressions (regex)* are a great tool for text analysis and building rule-based system. *Regex* is a set of characters or a pattern that is used to match and find substrings in the text.
- For example, "[A-Z0-9a-z._%+~]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,4}" is used to find all email IDs in the text.
- *regex* are a great way to incorporate domain knowledge in NLP systems. For example, a range of product codes that maps to brand names can be used with regex in building an automatically identifying the complaints of the products.

Heuristics-Based NLP (Contd...)

- *Context Free Grammar* (CFG) are another resources used in rule-based system that is used to model natural languages.
- CFGs can be used to capture more complex and hierarchical information that a regex may not.
- Parsers like CKY and the Earley parser allows parsing of all kinds of CFGs

Building blocks of Heuristic-Based NLP

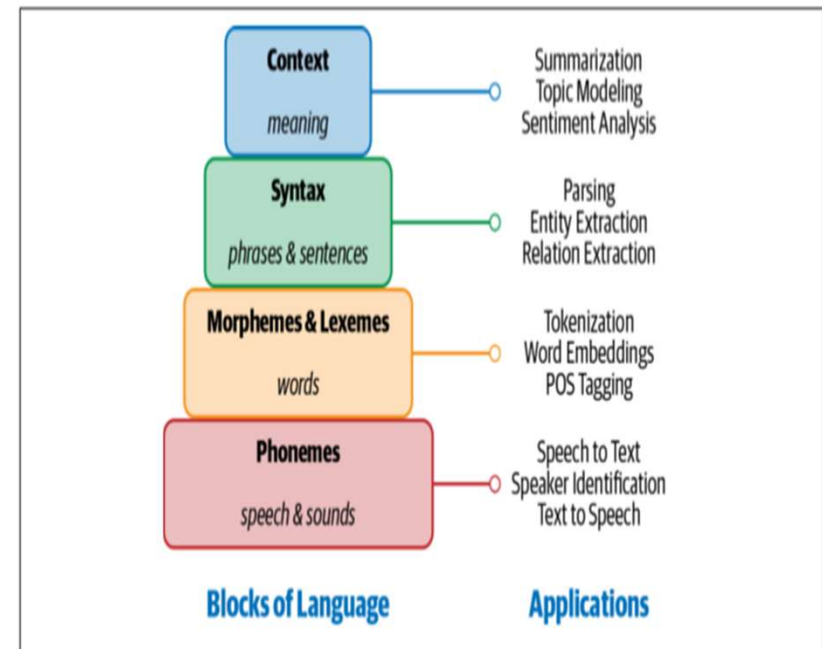
- Following building blocks of language are useful in designing rule-based systems:

1. *Phonemes*: Phonemes are the smallest units of sound in a language.

- They may or may not have any meaning by themselves but can induce meaning when uttered in combination.
- These are useful in applications such as speech recognition, speech-to-text transcription, and text-to-speech conversion.

2. *Morphemes and Lexemes*: These are words or parts of words which may or may not be meaningful.

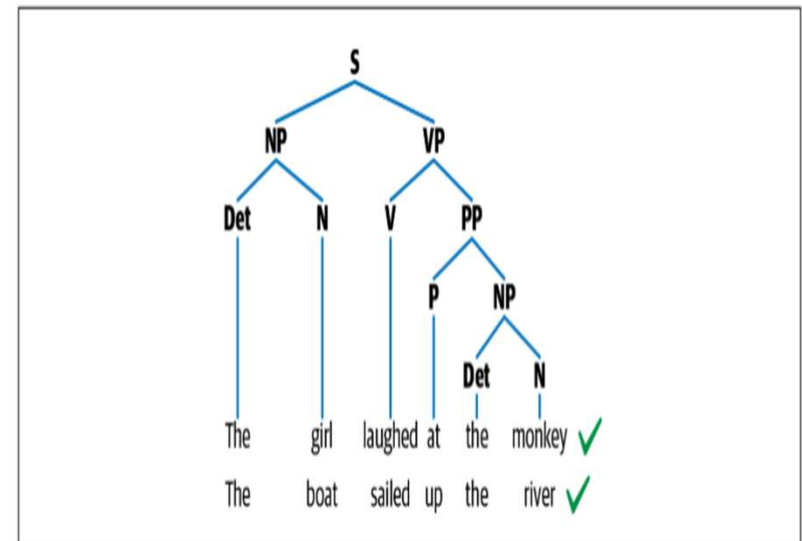
- Morphological analysis* is the building blocks of number of NLP tasks such as tokenization, stemming, learning word embeddings.



Building blocks of language and their applications

Building blocks of Heuristic-Based NLP (Contd...)

3. *Syntax*: Syntax is a set of rules to construct grammatically correct sentences out of words and phrases in language.
- Syntax guides the fundamental tasks of language processing such as parsing.
 - Entity extraction and relation extraction are some NLP tasks that are build on the knowledge of parsing.



Syntactic structure of two syntactically similar sentences

Building blocks of Heuristic-Based NLP (Contd...)

4. *Context*: Context is how various parts of a language come together to convey a particular meaning.
- Context includes meanings of words and phrases, long-term references, world knowledge, and common sense knowledge.
 - Context is generally composed of *semantics and pragmatics*. *Semantics* is the direct meaning of the words, phrases, and sentences without external context. *Pragmatics* add world knowledge and external context of the conversation to infer the implied meaning.
 - *Complex NLP tasks such as sarcasm detection, summarization, topic modeling are some tasks that uses context heavily.*

Limitations of Heuristic NLP

1. **Lack of Adaptability:** Heuristic NLP systems often struggle to adapt to new languages, domains, or contexts. Since they are designed based on predefined rules, they may not handle variations or changes effectively.
2. **Limited Generalization:** Heuristic approaches may not generalize well to handle a wide range of inputs. They are often tailored to specific use cases and may not perform well when faced with diverse or unexpected language patterns.
3. **Maintenance Challenges:** Maintaining and updating rule-based systems can be labor-intensive. As languages evolve and new expressions emerge, the rules need constant modification, making it difficult to keep the system up-to-date.
4. **Difficulty in Handling Ambiguity:** Natural language is inherently ambiguous, and heuristics may struggle to handle multiple interpretations of a given sentence. This can result in inaccurate or inconsistent results.

Limitations of Heuristic NLP (Contd.....)

5. **Scalability Issues:** Heuristic approaches may become impractical when dealing with large datasets or extensive language corpora. The manual creation and management of rules become increasingly complex and resource-intensive.
6. **Dependency on Linguistic Expertise:** Developing effective heuristics often requires in-depth linguistic expertise. This reliance on linguistic knowledge can make it challenging to create robust systems without the involvement of experts.
7. **Inability to Capture Semantics:** Heuristic approaches may not effectively capture the semantics and contextual nuances present in natural language. Understanding meaning beyond surface-level patterns can be challenging without the ability to learn from data.
8. **Limited Error Handling:** Heuristic systems may struggle to gracefully handle errors or unexpected inputs. Since they are built on predefined rules, they might produce unreliable results when faced with input that deviates from these rules.

Machine Learning- Based NLP

- Machine learning (ML) approaches have become increasingly popular for NLP tasks due to their ability to automatically learn patterns and relationships from data.

Some common machine learning approaches used in NLP:

1. Supervised Learning:

- **Text Classification:** Assigning predefined labels or categories to text documents. Examples include spam detection, sentiment analysis, and topic categorization.
- **Named Entity Recognition (NER):** Identifying and classifying entities (e.g., names of people, organizations, locations) in text.

2. Unsupervised Learning:

- **Clustering:** Grouping similar documents or words together based on patterns or similarities without predefined labels. Topic modeling, such as Latent Dirichlet Allocation (LDA), is an example.
- **Word Embeddings:** Representing words as dense vectors in a continuous vector space. Word2Vec, GloVe, and FastText are popular techniques.

Machine Learning- Based NLP (Contd.....)

3. Semi-Supervised Learning:

- Combining elements of both supervised and unsupervised learning, this approach leverages both labeled and unlabeled data. It is particularly useful when obtaining large labeled datasets is challenging.

4. Sequence Labeling:

- **Part-of-Speech Tagging (POS):** Assigning grammatical categories (nouns, verbs, adjectives, etc.) to each word in a sentence.
- **Chunking:** Identifying and classifying short phrases or "chunks" of information in a sentence.

Deep Learning

- In the last few years, there have been a huge surge in using neural networks to deal with complex, unstructured data.
- Language is inherently complex and unstructured. Therefore, we need better representation and learning capability to understand and solve language tasks.
- Some popular deep neural architectures that have been used in NLP are:
 1. **Recurrent Neural Networks (RNNs):**
 - Effective for tasks involving sequential data, RNNs process information in a sequential manner
 - used for tasks like machine translation, text summarization, and language modeling.

Deep Learning (Contd.....)

2. **Long Short-Term Memory Networks (LSTMs) and Gated Recurrent Units (GRUs):**
 - These are specialized types of RNNs designed to address the vanishing gradient problem, enabling better learning of long-range dependencies in sequential data.
3. **Transformer Models:**
 - **Attention Mechanism:** Transformers, introduced by models like BERT, GPT, and T5, use self-attention mechanisms to capture contextual information in a more effective way.
 - These models have achieved state-of-the-art performance on various NLP tasks, such as language understanding, translation, and summarization.
4. **Transfer Learning:**
 - Leveraging pre-trained models on large datasets and fine-tuning them for specific tasks. BERT and GPT are examples of models pre-trained on large corpora and then fine-tuned for specific downstream tasks.

Deep Learning (Contd.....)

5. Ensemble Learning:

- Combining predictions from multiple models to improve overall performance. Ensemble methods, such as bagging and boosting, are applied to NLP tasks for enhanced accuracy and robustness.

6. Neural Machine Translation (NMT):

- Using neural networks, particularly sequence-to-sequence models, for machine translation tasks where the goal is to translate text from one language to another.

DL not yet the silver bullet for NLP

While deep learning has shown remarkable success in various NLP tasks, it is not considered a silver bullet for several reasons:

Data Requirements:

- Deep learning models, especially large-scale ones like BERT and GPT, often require massive amounts of labeled data for pre-training.
- Obtaining such datasets can be challenging, particularly for languages with limited resources or for specialized domains.

Computational Resources:

- Training and fine-tuning deep learning models can be computationally expensive.
- Training large transformer models may require powerful hardware like GPUs or TPUs and extensive computational resources, making them less accessible for smaller organizations or research projects.

DL not yet the silver bullet for NLP

Interpretability:

- Deep learning models, especially those with many layers, are often considered black boxes, making it difficult to interpret their decision-making processes.
- Interpretability is crucial in applications like healthcare and finance, where understanding model decisions is essential.

Domain Adaptability:

- Pre-trained models might not easily adapt to specific domains or tasks without extensive fine-tuning.
- If the pre-training data is vastly different from the target domain, the model may not generalize well.

Multimodal Challenges:

- Integrating information from multiple modalities, such as text and images, remains a challenge for deep learning models. While progress is being made, it is an area that requires further research.